Gyanaranjan Mishra, Data Scientist

Scarborough, ON M1K 1R6 | gyanaranjanmishra06@gmail.com 647-906-6741 | LinkedIn | GitHub

SUMMARY

Materials Scientist turned Data Scientist with 13+ years in R&D at Tata Steel, specializing in predictive modeling, large-scale data analysis, and ML-driven research. Recently completed a PhD incorporating data-driven modeling and multi-source analytics. Built progressive ML expertise since 2017–18, with hands-on experience in Python, SQL, scikit-learn, and TensorFlow - building and evaluating models on datasets exceeding 5M records, improving prediction accuracy by 22%.

Now transitioning into a dedicated Data Science role, combining deep domain expertise, end-to-end ML workflow experience, and a continuous investment in modern cloud and AI tooling - with a track record of translating complex operational data into measurable business impact.

SKILLS

Programming & Data: Python (Pandas, NumPy, Scikit-learn, TensorFlow, Keras), SQL, Git, AWS

Machine Learning & Statistics: Regression, Classification, Clustering, Random Forest, XGBoost, SVM, Neural Networks, NLP (Word2Vec, NLTK), SMOTE, Hyperparameter Tuning, Cross-Validation, ROC-AUC, F1, Bias-Variance Trade-off, Time-Series & Forecasting

Analytics &Visualization: Exploratory Data Analysis (EDA), Matplotlib, Seaborn, Plotly, Power BI (Working Knowledge), Streamlit, Dashboard & Reporting Automation, Model Serialization (Pickle)

Domain & Systems: Process Optimization, Experimental Design, Manufacturing Systems, Project Management, Intellectual Property (IP), Quality Control (QC), Quality Assurance (QA)

RELEVANT EXPERIENCE

VISITING RESEARCHER - ADVANCED MATERIALS PROCESSING **Apr 2022 – Sep 2025**
CanmetMATERIALS | Natural Resources Canada (concurrent with PhD)

- Led PhD research partially focused on predictive modeling and deep learning for materials performance, resulting in peer-reviewed publication and transferable analytics frameworks
- Designed analytics pipelines for data cleaning, feature engineering, and exploratory analysis, improving dataset reliability by ~25%
- Developed predictive models (XGBoost, Random Forest, Neural Networks) achieving ROC-AUC > 0.85 for material performance evaluation
- Delivered technical and executive-level reports translating model outputs into actionable recommendations for industry partners
- Prepared market and production intelligence reports supporting strategic planning
- Collaborated with researchers, engineers, and external partners to translate analytical findings into development decisions

PRINCIPAL RESEARCHER - NEW PRODUCT DEVELOPMENT **Oct 2018 - Aug 2021**
Research and Development | Tata Steel Ltd, India

- Led product development programs supported by statistical analysis and machine learning-driven optimizations to enhance product quality, consistency, and manufacturing robustness supporting high value manufacturing operations
- Designed Python and MATLAB workflows to analyze high-volume experimental and production datasets to aid in product development
- Built end-to-end machine learning workflows for quality prediction and process optimization, reducing trial-and-error experimentation by ~20%
- Implemented regression, ensemble models, and neural networks to forecast mechanical properties of newly developed products for different market segments
- Automated performance dashboards and reporting pipelines, reducing manual reporting time by ~ 40%
- Collaborated with production, IT, and quality teams to embed analytics into operational decision-making
- Developed automated reporting templates of project charters and risk assessment reports for management review
- Applied root cause analysis and trend detection to diagnose quality deviations
- Contributed towards implementing large scale Industry 4.0 transformation across Tata Steel India
- Mentored junior researchers and coordinated cross-functional development teams

RESEARCHER - NEW PRODUCT DEVELOPMENT **Jul 2012 - Sep 2018**
Research and Development | Tata Steel Ltd, India

- Initiated large-scale analysis of process, quality, and metallurgical datasets to effectively reduce new product development life cycle by ~ 15%
- Developed regression and empirical models for alloy design providing cost effective solutions during steel product development
- Developed ML models to support operations teams with data-backed recommendations, improving operational effectiveness and reduced unit down time
- Performed multivariate analysis on operations and metallurgical data to identify key drivers of product performance variability
- Established standardized data collection protocols at level 2 of manufacturing units, enabling faster root cause analysis in case of exceptions
- Collaborated with production, maintenance, and quality teams to align project objectives with organizational goals
- Provided data driven insights to higher management to improve operational efficiency, reduce unit downtime and boost productivity

SELECTED PROJECTS

- **Deformation Stress Modeling (PhD) | Link :** Built two deep learning models using TensorFlow and Keras on multi-source experimental datasets exceeding 200K samples to predict high-temperature flow stress of electrical steels (1.3% and 3.2% Si variants), achieving ~25% improvement in prediction accuracy over conventional empirical models. Deployed as an interactive Streamlit web application.
- **Imbalanced Classification - Kick Car Prediction (Data Analytics Course) | Link :** Developed an end-to-end ML pipeline on 70K+ records with 30+ features to predict problematic used car purchases at auction. Applied XGBoost and Random Forest classifiers with SMOTE for class imbalance handling and ROC-AUC optimization. Deployed as a Streamlit app with serialized model artifacts.
- **Materials Performance Optimization (Tata Steel) :** Developed Regression and Classification based Machine Learning models to predict alloy compositions processing variables and material defects based on historical level 2 automation data, reducing product development cycle and QA time.
- **Quora Question Pair Similarity (PLP) | Link :** Built an NLP pipeline combining Word2Vec (Gensim) embeddings with feature engineering including fuzzy matching (FuzzyWuzzy), token overlap, and length-based features. Trained and optimized a Random Forest classifier to detect semantically similar question pairs, achieving ~90% accuracy. Deployed via Streamlit with automated model loading from Google Drive.

LICENSES & CERTIFICATIONS

- AWS Certified AI Practitioner — Amazon Web Services (AWS) **Apr 2026 - Apr 2029**
- SQL (Basic) — HackerRank **Apr 2026**
- Machine Learning Specialization — DeepLearning.AI, Stanford University **Oct 2023**
- Unsupervised Learning, Recommenders, Reinforcement Learning — DeepLearning.AI, Stanford University **Oct 2023**
- Supervised Machine Learning: Regression and Classification — DeepLearning.AI, Stanford University **Aug 2023**

PUBLICATIONS & PATENTS

- 15+ Peer-Reviewed publications in Materials Research
- 8+ Granted patents in Manufacturing Technology
- Full list of publications | Link

AWARDS & ACHIEVEMENTS

- RNB Graduate Merit Award - UNB, Canada **2021 - 2025**
- NBIF STEM & Social Innovation Award - UNB, Canada **2021 | 2022**
- Best Innovative Project Award - Tata Steel ltd, India **2016 | 2017**
- Institute Silver Medal - NIT Rourkela, India **2009**

EDUCATION

PhD, Mechanical Engineering | UNB, Canada | GPA - 4.2/4.3 **Sep 2021 - Dec 2025**

MTech, Metallurgical Engineering and Materials science | IIT Bombay, India | GPA - 8.76/10 **Jul 2010 - Jun 2012**

BTech, Metallurgical and Materials Engineering | NIT Rourkela, India | GPA - 9.15/10 **Jul 2005 - Apr 2009**